

SEPEHR NOEY KORDKANDI

✉ sepehr.nk.81@gmail.com · 🌐 sepehrnoey.github.io · in Sepehr Noey · 🐙 SepehrNoey

Education

B.Sc in Computer Engineering (Major in Artificial Intelligence) Sep. 2020 - Feb. 2025
Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

- **GPA:** 3.92/4
- **GPA (Last Two Years):** 4/4

Research Interests

- Computer Vision
- Vision-Language
- Machine Learning
- Multimodal Learning

Publications

[1] Alireza Salehi, Sepehr Noey, Sahand Noey, Mohammad Sabokrou. "Improving Anomaly Detection Using Self and Cross Consistency Between Images" (*In Preparation*)

Research Experience

B.Sc. Thesis, Amirkabir University of Technology, Tehran, Iran Sep. 2024 - Sep. 2025
Supervisor: *Dr. Mohammad Rahmati*

- Thesis Title: "Distillation of a Diffusion Model for Video Generation"
- Learned motion consistency from videos and frame quality enhancement from image datasets.
- Utilized Latent DMs and Image Discriminators in PyTorch and Hugging Face.

Research Intern, *Institute For Research In Fundamental Sciences*, [GitHub] July 2023 - Sep. 2023

- Fine-tuned a transformer-based model (mt5) on paired datasets for Persian grammar correction.
- Utilized Seq2Seq Learning and LoRA technique in PyTorch and Hugging Face.

Selected Projects

Landscape Image Generation Using Diffusion Models [GitHub] Spring 2024

- Implemented an image diffusion model for landscape image generation of size 32x32.
- Involved concepts: DDPM, U-Net architecture, and Attention.

Image Captioning Using Vision Transformers [GitHub] Spring 2024

- Implemented a vision encoder and text decoder to caption input images.
- Involved concepts: Sinusoidal Embeddings, and ViTs.

Data Mining Course Projects [GitHub] Spring 2024

- Used classification algorithms like XGBoost, SVM, KNN, Logistic Regression and Decision Trees.
- Used Linear and Polynomial Regression for estimation, and K-Means for clustering tasks.
- Used PCA for Dimensionality Reduction and SMOTE for over-sampling

Information Retrieval System [GitHub] Winter 2023

- Implemented Boolean and Ranked retrieval systems for Persian texts
- Concepts: Data cleaning, preprocessing, indexing, and efficient retrieval

ANN Framework From Scratch for Handwritten Digit Recognition [GitHub] Fall 2023

- Implemented a framework from scratch for training Fully-Connected Neural Networks (FCNN).
- Involved concepts: Neural networks, SGD, Adam, MSE, Cross-Entropy.
- Trained networks for classification tasks such as handwritten digit recognition.

Classic AI Projects [GitHub] Fall 2022

- Associated projects of Artificial Intelligence course in UC, Berkeley (**Pacman**).
- Implemented multiple search algorithms in AI including BFS, DFS, UCS, Greedy, and A*.
- Involved concepts like CSPs, bf MDPs, Q-Learning, Reinforcement Learning, and Bayesian Nets.

Teaching Experience

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

Head Teaching Assistant

- Microprocessor and Assembly Language *Dr. Hamed Farbeh* Spring 2023-2024

Teaching Assistant

- Principles of Cloud Computing *Dr. Seyyed Ahmad Javadi* Fall 2024
- Software Engineering II *Dr. Faezeh Gohari* Fall 2024
- Operating Systems (Lab) *Dr. Hamid R. Zarandi* Spring 2024
- Computer Networks *Dr. Fatemeh Ziaetabar* Spring 2024
- Computer Networks *Dr. Masoud Sabaei* Fall 2023-2024
- Applied Linear Algebra *Dr. Maryam Amirmazlaghani* Spring 2023
- Advanced Programming *Dr. S. Roostaei* Fall 2022
- Fund. of Computers and Programming *Parham Alvani* Spring 2022

Selected Courses

Major Related Courses

- Artificial Intelligence: 17.6/20 (A) ○ Computational Intelligence: 20/20 (A+)
- Research and Technical Presentation: 18.72/20 (A+) ○ Data Mining: 19.1/20 (A+)
- Information Retrieval: 20/20 (A+) ○ Algorithm Design: 18.25/20 (A+)

Online Courses

- DL for Computer Vision (**University of Michigan**) ○ Machine Learning (**Stanford University**)

Technical Skills & Languages

Programming Languages: Python, Java, C, Go, C#, JavaScript

Libraries & Frameworks: PyTorch, TensorFlow, OpenCV, Hugging Face, NumPy, scikit-learn

Tools: L^AT_EX, Linux, Git, Docker, Jupyter, Kaggle, VS Code

Languages:

- **English:** TOEFL iBT – Total: 103 (Reading: 30, Listening: 29, Speaking: 21, Writing: 23)
- **Persian:** Native

Honors & Awards

- Ranked among the **top students**, Amirkabir University of Technology 2023
- Awarded **Tuition-Waiver**, Amirkabir University of Technology 2020
- Ranked in the **top 0.2%** among 155,250 participants, **Iranian University Entrance Exam** 2020
- Awarded 2nd place in the mathematics competition of the NOET. 2019

References

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

- *Dr. Hamed Farbeh*, Assistant Professor, Computer Engineering Department
Email: farbeh@aut.ac.ir
- *Dr. Masoud Sabaei*, Associate Professor, Computer Engineering Department
Email: sabaei@aut.ac.ir
- *Dr. Alireza Bagheri*, Associate Professor, Computer Engineering Department
Email: ar_bagheri@aut.ac.ir